



CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: Research Assistant Platform

Group No: 10, CSE471 Lab Section: 10, Summer 2026		
SL	ID	Name
1	22201842	Md. Rafiul Islam
2	23101272	Reaz Ahmed Ramim
3	22201358	Wasi Farabi
4	23101294	Siffat Ara Naina

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Project Overview

The Research Assistant Platform is a web-based system that supports students and researchers across the full research lifecycle - from organizing and reading papers, to identifying research gaps, to managing collaborative research projects. The system supports four user roles: Researcher, Team Member, Supervisor, and Admin. It integrates meaningful AI tooling (summarization, insight extraction, gap analysis, an interactive paper-chat assistant, and title generation) rather than adding AI for its own sake, and gathers the heavier AI features behind a Free / Premium subscription model enforced through a monthly usage quota.

Tech Stack:

- **Language:** Python is used for both the frontend and backend layers.
- **Backend Framework:** FastAPI handles asynchronous REST API endpoints and data processing.
- **Frontend Framework:** Flask manages web page routing and renders Jinja2 templates.
- **Styling:** Tailwind CSS provides responsive, utility-first UI styling directly in HTML.
- **Database:** PostgreSQL stores relational data. It uses the pgvector extension for embeddings.
- **ORM:** SQLAlchemy manages database queries and object-relational mapping.
- **Background Processing:** Celery and Redis handle heavy PDF parsing, chunking, and embedding tasks.
- **Authentication:** JWT-based secure token management.
- **Deployment:** Hosted on Render or Railway (Backend) and Vercel or similar (Frontend).

External APIs:

- **Google Gemini API (gemini-2.0-flash):** Powers text summarization, insight extraction, gap analysis, question generation, and paper chat.
- **SSLCommerz / Stripe API:** Processes premium subscription transactions via secure payment gateways.
- **SMTP / Gmail API:** Dispatches automated email notifications for project milestones and system deadlines.

User Roles

1. **Researcher:** The primary user - uploads and organizes papers, takes notes, uses the AI research tools, and creates/manages research projects.
2. **Team Member:** A collaborator invited into a research project - shares tasks, papers, and discussion within that project's workspace.
3. **Supervisor:** Reviews a project's progress, leaves feedback on submitted work, and approves or requests revisions on milestones.
4. **Admin:** Platform manager - monitors overall usage, manages subscription tiers, and oversees AI request quotas and costs across users.

Functional Requirements

SL	Common Workflows
1	Registration and Login System: Users register by providing name, email, and password, and select a role (Researcher or Team Member); Supervisors are invited into a specific project by an existing member rather than self-registering. Once registered, users log in securely via JWT-based authentication to reach their personalized dashboard.
2	Admin Role & Access Management: An authorized Admin signs into a dedicated panel to monitor platform-wide activity - total users, AI request volume, and active subscriptions. The Admin can adjust subscription tier limits, manually resolve payment disputes, and review flagged or unusually high AI usage.

Module 1	
Member	Feature Description
1	Dashboard Overview — Users see a summary card showing total papers uploaded, total projects joined, number of completed papers, and their current subscription plan (Free/Premium).
2	Paper Upload & Library — Researchers upload PDFs to a personal library. Papers can be filtered and organized by topic, year, author, or research area. Custom tags can be added to any paper.
3	Paper Reading Progress Tracker — Tracks each paper's reading status (Unread / Reading / Completed), read percentage (0–100%), and total time spent reading. Visible from the library list view.
4	Research Project Creation — Researchers create named projects (e.g. "Federated Learning for Pest Classification") with a title and description. The creator becomes the project owner automatically.

Module 2	
Member	Feature Description
1	Paper Overview Panel — A dedicated page for a single paper showing its extracted abstract (pulled from the PDF using section-header detection, not a blind slice), reading status, link to a project, and a notes section for paper-specific annotations tied to page numbers.
2	Bookmarking System — Researchers bookmark important papers for quick access. A dedicated "Saved Papers" view shows all bookmarked papers. Bookmarks can be added or removed from the paper detail page.
3	Project Membership & Invitations — Project owners invite other users by email, assigning them a role of Collaborator or Supervisor. The project detail page shows all current members and their roles.

4	Task Management — Project members create and assign research-specific tasks (dataset collection, model training, paper writing, etc.) with status tracking (To-Do / In Progress / Review / Done) and optional due dates.
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Module 3	
Member	Feature Description
1	AI Paper Summarization — Generates a structured AI summary of an uploaded paper covering three sections: abstract summary, methodology summary, and key findings. Uses the Gemini API with the paper's extracted text as context.
2	Key Insight Extraction — Extracts a structured breakdown from a paper: contributions, advantages, limitations, and future work — presented as bullet lists rather than a single freeform summary.
3	Research Gap Finder — User selects 2 or more papers. The system synthesizes their limitations and future-work sections via Gemini to surface recurring unsolved problems and potential research directions.
4	Research Question Generator — Given a topic (e.g. "Traffic congestion in Dhaka"), generates candidate research questions, measurable objectives, and testable hypotheses the researcher can refine and save to a project.

Module 4	
Member	Feature Description
1	Citation Generator — Generates citations in APA, IEEE, and ACM formats from a paper's stored metadata (title, authors, year) using deterministic rule-based templates — no AI involved. One-click copy for each format.

2	Literature Comparison Matrix — User selects multiple papers. The system auto-compiles a structured comparison table showing Paper Title, Dataset Used, Method/Model, and Key Result/Accuracy — using Gemini to extract fields from paper text.
3	Supervisor Milestone Approval — Project members submit milestones (Proposal, Mid-Defense, Final Paper, Defense) for supervisor review. Supervisors approve or request revisions with written feedback. Approvals are logged with the supervisor's ID and timestamp.
4	Collaboration Discussion Board — A per-project threaded discussion board where team members and the supervisor can post updates and comments. Messages are timestamped and attributed to the sender by name and role.

Additional Features

#	Feature Description
1	AI Chat with Paper — Users ask natural-language questions about a specific paper (e.g. "What dataset did this paper use?"). The system answers directly from the paper's extracted text using the Gemini API.
2	AI Title Generator — Generates 5 candidate paper or thesis titles from a short project description or abstract, using the Gemini API.
3	Related Paper Recommendations — Suggests similar papers from the user's own library based on keyword overlap between paper titles and tags, computed dynamically.
4	Research Journal — A private, dated log where researchers record daily research progress, notes from meetings, and updates on papers and projects.
5	Research Timeline Tracker — Visual timeline of a project's key milestones (Proposal, Mid-Defense, Final Paper, Defense) showing completed, pending-review, and upcoming stages in chronological order.
6	Meeting Scheduler — Project members request specific dates and times for supervisor consultations. The supervisor can view, accept, or propose an alternative time.